

BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet

Electron Configuration

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: shells, subshells, Aufbau, orbital boxes and ion configurations.

A Retrieval

1. State the maximum number of electrons in the first, second and third shells for the first 20 elements as used in this course.

2. How many electrons can an s subshell and a p subshell hold?

3. What does the group number tell you about a Group 1 or Group 7 atom?

B Knowledge

4. Write shell configurations for N, Mg, Cl and K.

5. Write full subshell configurations for O, Al and Ca. Use superscripts, e.g. $1s^2 2s^2 2p^4$.

Orbital box diagrams

Each box is one orbital. Arrows are electrons of opposite spin.

Helium 1s²

Pauli: two electrons in one orbital must have opposite spin.

Nitrogen 2p³

Hund: one electron in each equal-energy orbital before pairing.

Oxygen 2p⁴

The fourth 2p electron pairs in one box.

6. On the orbital-box idea above, draw boxes and arrows for nitrogen 2p³ and oxygen 2p⁴. State Hund's rule in one sentence.

C Application

7. Write the configuration of O²⁻, Al³⁺ and Cl⁻ in both shell and subshell form.

8. A student writes calcium as 2,8,10. Correct the configuration and explain the error.

9. Explain, using electrons, why magnesium forms a 2+ ion.

D Exam-style practice

10. Write the electron configuration of Mg^{2+} and explain why the ion is stable. [3]

11. Explain why oxygen typically forms a $2-$ ion. [3]

12. Identify the block (s, p or d) for sodium, fluorine and iron, and justify one of them. [3]

E Challenge

13. Write [Ne] shorthand configurations for phosphorus and for P^{3-} . Explain the relationship between them.
